

Statistical Language Modelling of Disfluent Speech

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Dysfluent Speech?

A speech that is not smooth or flowing, and is interrupted by breaks, repetitions or pauses.

Why is dysfluency detection important?

Dysfluency detection is important as stuttering directly affects the communication and confidence. Currently, diagnosis and therapy rely heavily on manual assessment by experts, which is time-consuming and subjective. Automated systems can assist therapists by providing scalable and objective evaluation tools.

Technical Aspects

From a technical perspective, most prior research prioritizes acoustic features. However, lexical modeling remains underexplored. Since modern ASR systems like Whisper by OpenAI may normalize repetitions, it is important to investigate whether lexical representations alone can capture dysfluency patterns. This helps us understand the limits of text-based modeling and build strong baselines.

Types of Disfluencies

- Sound Repetition
- Syllable Repetition
- Prolongation
- Block (Silent or Audible)
- Interjection (Filler)
- Phrase Repetition
- Cluttering

Work done so far (existing research)

- Limited research has been done on the lexical features for the stuttering related dysfluency detection models.
- Most existing systems prioritize acoustic modeling
- The lexical data is generated mainly using whisper, which is a weakly supervised model which may normalise pauses and repetitions.
- A significant research gap exists in robust lexical-only modeling of dysfluency patterns.

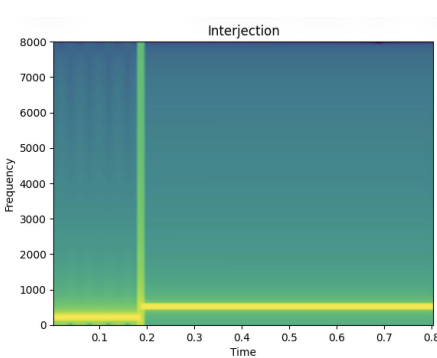
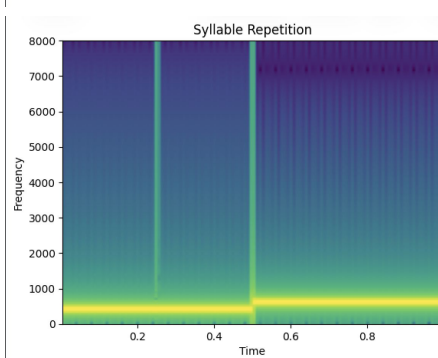
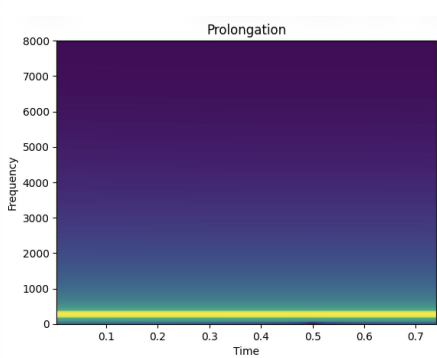
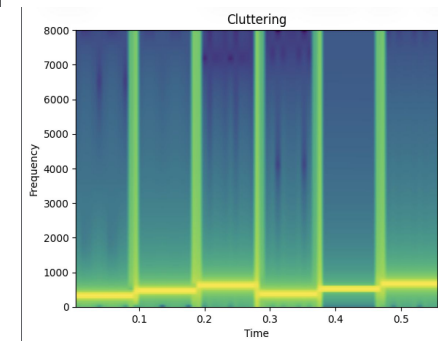
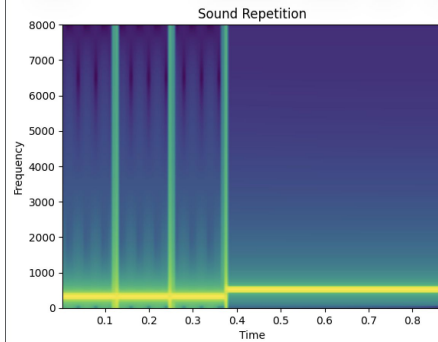
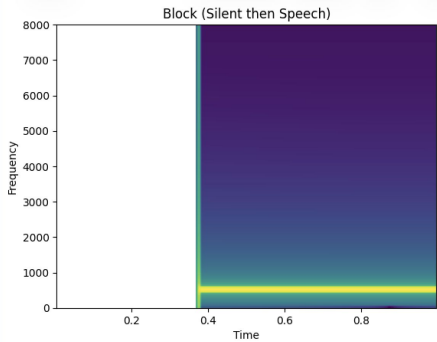
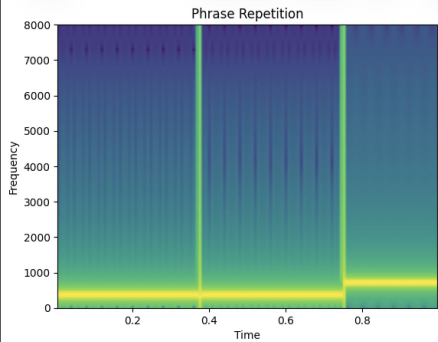
Problem Statement

Our goal is to make a model that can correctly categorise the transcripts for accurate dysfluency and also correct it to further refine whisper.

Understanding the Data

Firstly, we jumped right into the project by studying the research papers that gave us a great foundation to work upon.

We then, analysed the types of dysfluencies and got their spectro-graphical representation for better understanding.



Large Language Models for Dysfluency Detection in Stuttered Speech

This is a research paper published on September 2024, they have used a multimodel approach by incorporating both lexical and acoustic data samples, using whisper and wav2vec respectively. Feeding the output into a llama2 model. The task is reformulated as a generative language problem where dysfluency labels are produced as tokens rather than going with a classifying approach.

Wagner, M., et al., "Large Language Models for Dysfluency Detection in Stuttered Speech," in *Proceedings of Interspeech 2024*, September 2024.

Data Pre-Processing

We created our own synthetic data by implementing a pipeline, on the data set Librispeech.asr, which resulted in us having a total of 160,000 transcriptions.

N-Gram LM

Currently our focus is on the lexical features and so we are using the N-gram LM to capture the local repeating patterns and create a metric for it.

Repetition ratio = number of repeating tokens/total tokens

So, a higher ratio meaning a higher probability of having stuttering.

Thank you

This is the progress that we have made till today.